

Shawn Dsouza

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PROFILE SUMMARY

I am a behavioural ecologist with 8 years of research experience investigating how anthropogenic disturbance alters animal behaviour, trophic interactions, and ecosystem functioning. My work integrates meta-analysis, causal inference, agent-based models, and field experiments to quantify behavioural change and its ecological consequences. I have a strong publication record, experience leading interdisciplinary projects, and a proven ability to tackle complex ecological questions. I am seeking postdoctoral opportunities that combine theory, data synthesis, and empirical approaches to address questions in global change and behavioural ecology.

RESEARCH INTERESTS

I am interested in how anthropogenic disturbance reshapes natural systems by altering animal behaviour, trophic interactions and ecosystem functioning. My research focuses on understanding how animals perceive and respond to humans in their environment, and how these behavioural shifts cascade across trophic levels to influence ecosystem structure and function. I integrate multiple approaches including systematic reviews and meta-analyses, agent-based modelling, Bayesian inference, and field-based experiments to study behaviour across spatial and temporal scales. Going forward, I aim to develop predictive frameworks that combine behavioural theory with data-driven models to better anticipate ecological responses to global change, and to inform conservation and management under uncertainty.

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

September 2026 – Present

National Centre for Biological Sciences (NCBS), Bengaluru — CaFE Lab (PI: Dr. Meghna Krishnadas)

I work on long-term monitoring and plant trait data to determine how trait–environment interactions shape seedling survival and community assembly across environmental and disturbance gradients.

Research Affiliate

May 2022 – December 2024

Dakshin Foundation, Bangalore

I led the elasmobranch monitoring program in Malvan, Maharashtra, overseeing data collection across multiple landing centers. I also consulted various ongoing projects concerning marine systems and fisheries.

Research Assistant

June 2017 – February 2021

Dakshin Foundation, Bangalore

I studied the effects of fisheries on sea snake species' interactions using a combination of stable isotope analysis and fisheries dependent methods. I also managed the bycatch projects at the organisation as well as outreach and communication activities for the same.

EDUCATION

Ph.D. in Biology

2021 – 2026

Center for Ecological Sciences, Indian Institute of Science (IISc), Bangalore

Thesis: Impacts of human disturbance on animal behaviour and ecological system functioning.

I conducted research on human disturbance and its impact on natural systems, focusing on changes in animal behaviour and its implications for the functioning of ecological systems. I developed methodologies to assess and understand anthropogenic impacts on animal behaviour and ecology through various tools including meta-analysis of available literature, agent based models, remote underwater video surveys and predator model experiments. I have submitted my thesis and am currently awaiting reviews.

Advisor: Prof. Kartik Shanker.

M.Sc. in Biodiversity

2017

M.E.S. Abasaheb Garware College, Savitribai Phule Pune University

Thesis: Impacts of human disturbance on animal behaviour and ecological system functioning.

CGPA: 8.65/10

B.Sc. in Zoology

2015

*S.I.E.S. College of Arts, Science and Commerce, Mumbai University***GRANTS, AWARDS & FELLOWSHIPS**

- ISBE Travel Grant, International Society for Behavioural Ecology — \$1,600 (2026)
- ISBE Travel Grant, International Society for Behavioural Ecology — \$2,500 (2024)
- Prime Minister's Research Fellowship (PMRF), IISc Bangalore (2022)
- GATE Fellowship, IISc Bangalore (2021)
- Mangrove Foundation of Maharashtra Small Grant — ₹4,70,000 (2018)
- Rufford Foundation, UK Small Grant — ~£4,500 (2018)
- World Wildlife Fund (WWF) – India Small Grant — ₹80,000 (2017)

PUBLICATIONS

Peer-reviewed publications

- Dsouza, S., Shanker, K., & Thaker, M. (2025). Are human super-predators always super-scary? A meta-analysis of wild animal behavioural responses to human interactions. *Ecology Letters*. 28(12). e70287.
- Patre, M. V., Bora, G., Jakhete, S., Dsouza, S., & Shanker, K. (2026). Microplastic accumulation in marine organisms across trophic levels along the west coast of India. *Marine Pollution Bulletin*, 222, 118816.
- Ruidas, H., Bora, G., Dsouza, S., & Shanker, K. (2025). Assessing the impact of microplastics on gonadal health of the spadenose shark (*Scoliodon laticaudus*) on the west coast of India. *Marine Pollution Bulletin*, 212, 117464.
- Bora, G., Dsouza, S., & Shanker, K. (2024). Diet composition and variation in four commonly landed and threatened shark species in Maharashtra, India. *Regional Studies in Marine Science*, 74, 103531.
- Dsouza, S., & Padhye, A. (2022). A cross taxonomic comparison of bird and butterfly communities of Tamhini Wildlife Sanctuary across two decades. *Current Science*. 123(10). 1253-1258.
- Dsouza, S., Rao, C., Manoharakrishnan, M., & Shanker, K. (2021). Resource use and the impacts of fisheries on two sympatric sea snake species on the west coast of India. *Marine Biology*, 168(9), 1-11.
- Dsouza, S., & Rao, C. (2021). Demographics and reproductive biology of *Hydrophis schistosus* may make it more resilient to bycatch effects than other sea snakes. *Regional Studies in Marine Science*, 47, 101948.
- Rao, C., Dsouza, S., Gupta, T., Manoharakrishnan, M., & Lobo, A. S. (2021). Fisheries induced shift in sea snake community assemblages along the Konkan coast, India. *Aquatic Conservation: Marine and Freshwater Ecosystems*, 31(9), 2402-2411.
- Jojan, A., Dsouza, S., Mukerjee, P., Rao, C., & Shanker, K. (2018). Critiquing narrow critiques of convention on biological diversity. *Economic & Political Weekly*, 53(44), 17.

Preprints under review

- Dsouza, S., Joshi, A., Ahmad, S., Ahuja, B., & Shanker, K. (2025). Local extraction of predatory fish may initiate behaviourally mediated trophic cascades in coral reefs in the Andaman Islands. *BioRxiv*. 2025-11
- Dsouza, S., Joshi, A., & Shanker, K. (2025). Does the loss of piscivorous predators in a coral reef alter risk allocation in reef-associated fish? *BioRxiv*. 2025-11

Conference presentations and posters

- “Are human super-predators always super-scary? A meta-analysis of wild animal behavioural responses to human interactions.” Poster at GRS Predator–Prey Interactions Seminar 2026, Barga.
- “Local extraction of predatory fish may initiate behaviourally mediated trophic cascades in coral reefs in the Andaman Islands.” Poster at GRC Predator–Prey Interactions Meeting 2026, Barga.
- “Are super-predators always super-scary? A meta-analysis of wild animal behavioural responses to human interactions.” Talk at Behaviour 2025, Kolkata.
- “Fishes do fear the reaper: marine protected areas alter fish assemblages and behaviour.” Talk at Behaviour 2025, Kolkata.
- “Are superpredators also super-scary? A systematic review and meta-analysis of the effect of human interactions on animal behaviour.” Talk at ISBE Annual Meeting 2024, Melbourne.
- “Fishing for sea snakes.” Student talk at SCCS Cambridge 2019.
- “Can butterflies improve conservation?” Student poster at SCCS Bangalore 2017.

Popular articles

- Dsouza, S. 2020. Locked Down in a Fishing Town. The Bastion.
- Dsouza, S. 2019. Scaly Business - A Day in the Life of a Sea Snake Ecologist. Current Conservation. 13.2.
- Creasey, M. and Dsouza, S. 2019. Sailing into an Uncertain Future. Current Conservation. 13.3.

TECHNICAL SKILLS

I am eager to learn new tools and technologies to effectively solve challenges that come my way, driven by a strong affinity for computers and programming. I am proficient in R and Python, with a robust background in statistical analysis, including linear and non-linear regression modelling, classical hypothesis testing, and Bayesian analysis. My experience includes generalized linear and mixed models, zero-inflated and non-linear models, causal and counterfactual analyses, and reproducible computational workflows on Linux-based systems (Git, Docker, LaTeX). I have extensive field and experimental experience, including behavioural observations, predator cue experiments, Baited Remote Underwater Video Surveys (BRUVS), fisheries-dependent sampling, and stable isotope-based food-web analyses. I also have experience in grant writing, scientific synthesis, and communicating research to academic, policy, and public audiences.

PROFESSIONAL LEADERSHIP & SERVICE

Symposia

- “Global Change Impacts on Predator-Prey Ecology and Evolution” Symposium at GRS Predator–Prey Interactions Seminar 2026, Barga.
- “Responses and adaptations to human super-predators and anthropogenically modified environments” Symposium at Behaviour 2025, Kolkata. Co-chairs: Chayan Munshi and Maria Thaker.

Invited talks

- “Sea Snakes: Ecology and Threats.” Talk at Centre for Ecology and Conservation, University of Exeter (2019).

Memberships

- International Society of Behavioural Ecologists (ISBE) (Since 2023)
- Association for the Study of Animal Behaviour (ASAB) (Since 2022)

REFERENCES

Prof. Kartik Shanker — Center for Ecological Sciences, Indian Institute of Science, Bangalore. Email: kshanker@iisc.ac.in

